

VERMILION AGDM, AB, Canada

WMO: 712780

Lat: 53.3439N

Lon: 110.8819W

Elev: 2044

StdP: 13.64

Time Zone: -7.00 (NAM)

Period: 03-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-25.3	-19.4	-31.9	1.0	-25.2	-25.5	1.4	-19.1	22.1	21.1	19.6	16.0	4.1	150	0.662

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.9	83.7	64.0	80.2	62.5	77.1	61.4	67.4	78.7	65.2	76.1	63.2	73.8	8.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.2	93.7	73.7	60.9	86.3	70.9	59.0	80.4	68.1	33.0	79.3	31.3	75.7	29.7	74.1	74.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.7	18.1	15.9	-33.5	89.0	4.9	3.1	-37.0	91.3	-39.9	93.1	-42.6	94.9	-46.2	97.1
			-33.6	71.8	4.7	2.1	-37.0	73.3	-39.7	74.5	-42.4	75.7	-45.8	77.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	36.4	10.1	9.8	22.5	38.4	50.8	58.4	62.8	60.0	51.4	38.6	22.1	10.6	(6)
	DBStd	22.28	14.98	14.15	13.08	9.54	7.80	5.53	4.93	5.76	7.95	8.75	12.66	13.79	(7)
	HDD50	6172	1237	1127	852	360	86	7	0	3	76	364	838	1222	(8)
	HDD65	10490	1702	1547	1317	798	443	206	102	171	411	818	1288	1687	(9)
	CDD50	1221	0	0	0	11	110	260	397	314	118	11	0	0	(10)
	CDD65	64	0	0	0	0	2	9	34	17	2	0	0	0	(11)
	CDH74	1424	0	0	0	3	107	202	549	390	168	5	0	0	(12)
	CDH80	313	0	0	0	0	14	33	134	83	49	0	0	0	(13)
	WSAvg	7.1	7.2	6.7	7.5	8.6	8.2	7.0	6.0	5.9	6.6	7.5	7.3	6.6	(14)
	PrecAvg	16.4	0.7	0.5	0.6	1.1	1.7	2.9	3.1	2.3	1.4	0.7	0.6	0.7	(15)
(16) Precipitation	PrecMax	23.8	1.7	1.1	1.7	3.6	4.0	5.2	6.5	5.8	3.8	1.5	2.1	1.3	(16)
	PrecMin	8.8	0.2	0.2	0.2	0.1	0.2	0.6	0.9	0.4	0.1	0.1	0.1	0.3	(17)
	PrecStd	3.5	0.4	0.2	0.4	0.8	1.0	1.2	1.5	1.0	0.9	0.4	0.5	0.3	(18)
	PrecStd	3.5	0.4	0.2	0.4	0.8	1.0	1.2	1.5	1.0	0.9	0.4	0.5	0.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.0	42.0	53.7	72.2	81.9	83.6	88.3	86.3	86.1	73.1	54.9	42.8	(19)
		MCWB	36.2	36.2	42.9	51.8	56.7	62.8	67.4	64.3	61.2	55.0	43.3	36.6	(20)
	2%	DB	38.4	37.7	46.4	66.2	77.0	79.4	83.9	81.9	78.7	63.1	46.7	37.9	(21)
		MCWB	34.4	33.5	38.9	48.4	55.0	60.9	67.1	63.4	58.9	49.5	39.6	33.7	(22)
	5%	DB	35.4	34.1	41.9	60.2	72.3	75.7	79.8	78.5	72.9	57.7	41.3	34.2	(23)
		MCWB	32.3	31.3	36.4	45.4	53.2	59.1	65.0	62.8	56.7	46.3	36.0	30.8	(24)
	10%	DB	31.4	30.7	38.6	54.9	68.1	72.1	76.5	74.7	67.1	52.8	37.4	30.0	(25)
		MCWB	28.9	28.1	34.4	43.1	51.1	57.6	63.5	61.5	53.9	44.3	33.7	27.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.6	36.8	44.3	53.0	59.2	66.2	71.6	69.0	62.8	55.5	44.3	37.3	(27)
		MCDB	41.0	41.2	52.2	69.3	75.1	79.0	83.4	79.5	81.4	71.5	53.0	42.6	(28)
	2%	WB	34.6	34.0	39.6	49.3	56.5	63.3	68.5	66.4	59.9	51.2	40.0	34.0	(29)
		MCDB	38.0	37.0	45.8	64.9	72.3	74.9	80.0	77.1	76.1	60.3	45.9	37.6	(30)
	5%	WB	32.4	31.4	36.7	46.1	54.5	61.1	66.5	64.1	57.5	47.8	36.6	31.0	(31)
		MCDB	35.3	34.1	41.4	58.2	69.1	71.9	77.6	75.2	70.9	56.3	40.4	34.0	(32)
	10%	WB	28.9	28.1	34.6	43.6	52.5	59.2	64.5	62.2	54.8	44.9	33.9	27.6	(33)
		MCDB	31.4	30.6	38.3	53.8	66.0	69.0	74.5	73.0	65.4	52.4	37.2	30.1	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	18.1	19.5	18.5	21.0	24.9	21.6	22.9	24.0	24.3	20.5	16.6	16.9	(35)
		MCDBR	18.8	21.8	21.4	31.1	31.9	28.7	29.7	31.8	35.3	29.9	21.4	20.7	(36)
		MCWBR	15.6	18.1	15.2	16.2	14.4	13.2	15.1	15.7	17.0	17.5	14.7	17.1	(37)
	5% WB	MCDBR	18.8	21.2	20.3	29.0	28.4	24.6	26.7	27.8	32.3	28.3	19.7	20.4	(38)
		MCWBR	15.8	17.8	14.5	16.0	13.5	12.8	14.6	14.9	16.4	17.2	14.0	17.2	(39)
(40) Clear-Sky Solar Irradiance	taub		0.247	0.293	0.287	0.331	0.351	0.372	0.366	0.378	0.313	0.277	0.250	0.220	(40)
	taud		2.323	2.202	2.379	2.362	2.352	2.326	2.368	2.312	2.472	2.545	2.448	2.359	(41)
	Ebn at Noon		240	254	288	286	284	278	278	267	276	267	241	233	(42)
	Edh at Noon		18	28	29	34	36	38	36	36	27	20	16	14	(43)
(44) All-Sky Solar Radiation	RadAvg		316	651	1128	1490	1848	1888	1957	1596	1124	652	316	235	(44)
	RadStd		33	47	89	131	128	136	105	119	122	91	34	23	(45)

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (65 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page